

TECHNICAL REPORT · ROAD TO GREEN

# Rooftop Solar PV System
# Design Basis & Cost Estimate

Marriott Hotel – Mezzanine Roof – Phase 1

| | |
|---|---|
| Document No. | NILESUN POWER RF002 |
| Revision | B – Updated costing and financials |
| Date | 9 November 2026 |
| Client | Kampala Marriott Hotel |
| System capacity | 185.4 kWp DC / 160 kW AC, grid-tied |

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Document Control

| Rev | Date | Description | Prepared | Checked | Approved |
|---|---|---|---|---|---|
| A | 3/11/2026 | Version A | Mahil Fernando | Prasanna Dissanayake | Manjula Perera |

## Reference documents

| Reference | Title |
|---|---|
| D18102-DD-A-104 | Mezzanine floor plan incl. roof PV zones (architect) |
| PV-LAY-D18102-002 | PV layout with ballast structure, 10° tilt |
| EPJ-26110401 V1.0 | Mounting structure drawings (2P and 1P tables) |

## Contents

## Contents

Reference documents....2
Contents....2
1. Executive Summary....3
Key findings....3
Recommendation....3
2. Site and Design Basis....4
2.1 Site and load....4
3.Estimation PV Solar Generation....5
4. Solar Impact on Time-of-Use Tariff....5
3.PV Array Layout....6
4. Electrical Design....7
4.1 Configuration....7
4.3 Single-line arrangement....7
5 Cost Estimate (Bill of Quantities)....8
7. Financial Analysis....8
Tariff sensitivity....9
Annual cash flow (selected years, USD)....9
8. Expansion Options....10
9. Assumptions, Exclusions and Next Steps....10
9.1 Key assumptions....10
9.2 Exclusions....10

## 1. Executive Summary

This report sets out the design basis, layout, energy yield and budget cost for a grid-tied rooftop solar photovoltaic (PV) system on the mezzanine-level roof of the Marriott Hotel,

The system has no battery storage and supplies the hotel's own load behind the utility meter.

| 185.4 kWp | 291 MWh | USD 105k | 4.1 yrs |
|---|---|---|---|
| DC capacity (299 × 620 Wp) | Year-1 generation | Total project cost (0.56 USD/Wp) | Simple payback |

### Key findings

From Energy audit and feasibility study key findings.

| Item | Value |
|---|---|
| Roof area (mezzanine roof) | 1,150 m² (Available area for solar installation) |
| Plant capacity | 185.4 kWp |
| Modules | 299 × 620 Wp (13 rows of 23 solar modules) |
| Structure | Fixed-tilt ballast structure, 10° |
| Inverters | 2 × 80 kW (160 kW AC) |
| Hotel consumption | 981,901 kWh/month (UEDCL bill, Oct 2026) |
| Self-consumption | 100% (no export, no battery) |
| Year-1 generation | 291 MWh (2.5% of annual consumption) – PVsyst P50 |
| Avoided tariff | USD 0.0871/kWh (UGX 357/kWh shoulder tariff, as billed) |
| Year-1 savings | USD 25,343 |
| Total project cost | USD 104,661 |
| Simple payback | 4.1 years |
| Project IRR | 25.5% |
| NPV (25 years at 10%) | USD 153,091 |
| Cost of solar energy (LCOE) | 0.041 USD/kWh |
| Road to 500 kWp | 315 kWp, needs other roofs or car-park canopies |

*Table 1: Summary of Energy Audit Finding & Recommended PV Solar Installation*

### Recommendation

Phase 1: Proceed with the 185.4 kWp mezzanine roof system.

Phase 2: Proceed with 250.4 kWp the car park

## 2. Site and Design Basis

### 2.1 Site and load

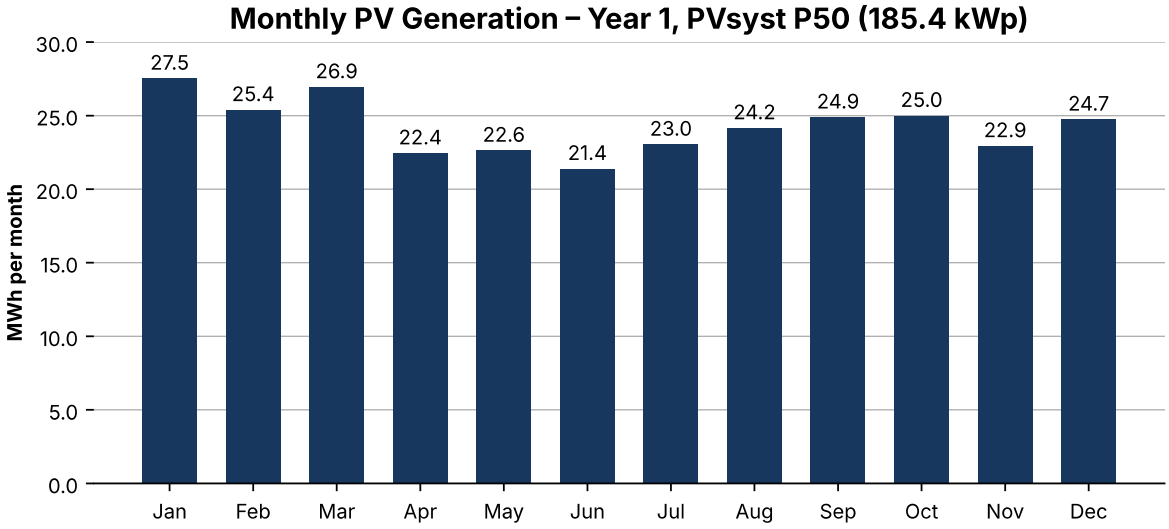
| | |
|---|---|
| **Facility** | Marriott Hotel –PV on the mezzanine-level roof |
| **Location** | 0°18'11.34"N, 32°35'13.76"E |
| **Roof** | Flat reinforced-concrete slab, single level, with parapet. |
| **Annual electricity consumption** | 11,782,812 kWh (October 2026 bill × 12) |
| **Average monthly energy consumption** | 981,901 kWh (UEDCL bill, 01/10–01/11/2026) |
| **Average / daytime demand** | ≈ 1,348 kW average in the shoulder band; maximum demand 1,836 kW |
| **Grid connection** | Behind-the-meter, LV main switchboard |
| **Energy storage** | None |




*Figure 1: PV module installation on the mezzanine roof ( AI Generated )*

## 3. Estimation of PV Solar Generation

- Year-1 generation 291,051 kWh, on average 24,254 kWh/month and 797 kWh/day
- 25-year generation 6,871 MWh*
- share of hotel uses 100 %, 2.5 % of annual consumption




*1.0 % degradation in year 1, then 0.4 %/yr

Source: PVsyst V8.1.8 simulation (P50, Meteonorm 9.0 weather, performance ratio 85.4 %, specific yield 1,570 kWh/kWp) – Annex 7. Output is highest from January to March and lowest in June.
The best month is about 29 % above the lowest. The hotel load is higher than the PV output in every hour, so all this energy is used on site. Figures are P50 as simulated, before shading and soiling losses (P90 ≈ 8 % lower).

## 4. Solar Impact on Time-of-Use Tariff

The hotel is billed on a time-of-use (TOU) tariff, so each kWh is worth a different amount depending on when it is used.

| Tariff period | Hours | Tariff UGX/kWh → USD/kWh (UGX 4,100/USD) |
|---|---|---|
| Off-peak | 00:00–06:00 | 286.5 → 0.0699 |
| Shoulder | 06:00–18:00 | 357.0 → 0.0871 |
| Peak | 18:00–24:00 | 428.8 → 0.1046 |

*Table 2: UEDCL tariff 30.2 Large Consumers (bill Oct 2026), tariffs as billed excl. VAT, at UGX 4,100/USD*

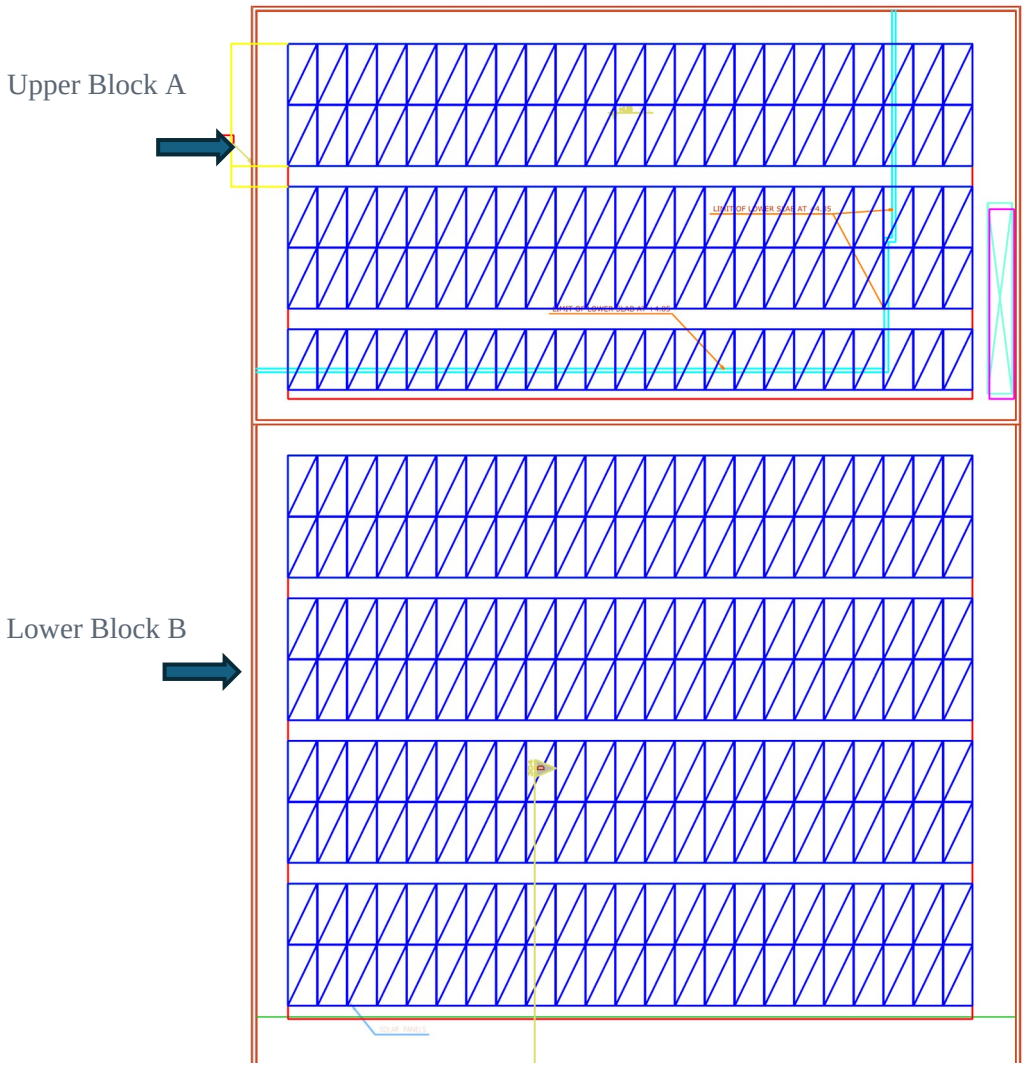
| Period | Hotel use (kWh/yr) | Solar (MWh/yr) | Share of solar | Grid after solar (kWh/yr) | Reduction | Cost before (USD/yr) | Cost after (USD/yr) | Saving (USD/yr) |
|---|---|---|---|---|---|---|---|---|
| Off-peak | 2,765 | 0.0 | 0.0% | 2,765 | 0.00% | 193,201 | 193,201 | 0 |
| Shoulder | 6,016 | 291.1 | 100.0% | 5,725 | 4.84% | 523,795 | 498,452 | 25,343 |
| Peak | 3,002 | 0.0 | 0.0% | 3,002 | 0.00% | 314,008 | 314,008 | 0 |
| **Total** | **11,783** | **291.1** | **100.0%** | **11,492** | **2.47%** | **1,031,004** | **1,005,661** | **25,343** |

*Table 2 : Energy reduction percentage & cost saving per year*

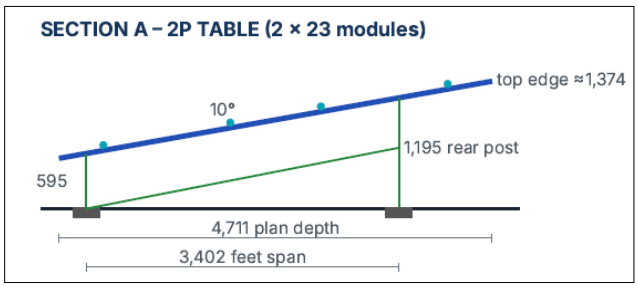
*Annual energy demand = UEDCL bill for October 2026 (981,901 kWh) × 12, split by tariff band as billed. All solar output (including the small output just after 18:00) is counted in the shoulder period. The demand charge (UGX 11,096/kW) is not reduced by solar.

## 5. PV Array Layout

Each row holds 23 modules on Enerack R80 rails 26,630 mm long, supported by 11 frames per table at 2,600 mm centers. The upper block carries two 2P tables and one 1P table; the lower block carries four 2P tables.




| Zone | Tables | Rows | Modules | kWp |
|---|---|---|---|---|
| Upper block A | 2 × 2P + 1 × 1P | 5 | 115 | 71.3 |
| Lower block B | 4 × 2P | 8 | 184 | 114.1 |
| **Total** | **7 tables** | **13** | **299** | **185.4** |



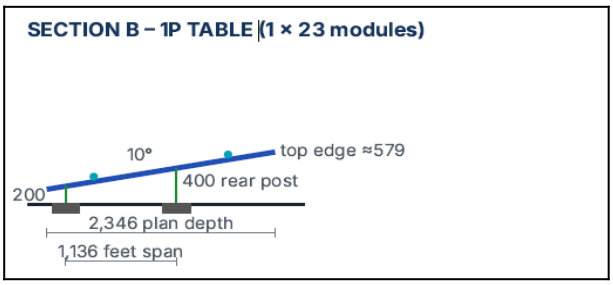



*Figure 3 : PV Layout and Table Structure details,*

## 6. Electrical Design

### 6.1 Configuration

| Item | Inverter INV-01 | Inverter INV-02 | Total |
|---|---|---|---|
| Rated AC power | 80 kW | 80 kW | 160 kW |
| Strings | 8 × 19 modules | 3 × 19 + 5 × 18 modules | 16 strings |
| Modules | 152 | 147 | 299 |
| DC capacity | 94.2 kWp | 91.1 kWp | 185.4 kWp |
| DC/AC ratio | 1.18 | 1.14 | 1.16 |

*Table 5: System Configuration & Inverter Capacity of each block*

### 6.2 Single-line arrangement

1. PV strings: 4 mm² double-insulated PV cable through the cable trays, via a DC isolator and a Type II DC SPD for every string, to the inverters on the roof in a shaded, ventilated location.
2. Inverters (400 V, 3-phase) → AC combiner panel with 2 × 160 A MCCB incomers, Type II AC SPD and 400 A roof isolator.
3. AC combiner panel → 4C × 240 mm² Al XLPE/SWA (≈ 200 m, from available spare stock) → existing Masterpact NW10 H1 (1000 A) solar incomer on the main LV switchboard; the cable's steel-wire armour serves as the protective earth (glands earthed).
4. Earthing: panel-to-panel bonding of the module frames, connected to the building earth and integrated with the lightning protection system.

| Component | Description | Size / Quantity |
|---|---|---|
| **Solar panels** | Generate DC electricity | 299 × 620 W = 185.4 kWp |
| **Inverters** | Convert DC power to AC power | 2 × 80 kW |
| **AC combiner panel** | Joins both inverter outputs on the roof | 1 panel, 400 A |
| **Main switchboard** | Where solar and grid power meet (hotel LV room) | Existing, 1000 A |
| **Utility grid (UEDCL)** | Supplies the rest of the hotel's electricity | Existing supply |

*Table 6 each item and in the block and Size/Quantity*

| Component | Description | Size / Quantity |
|---|---|---|
| **Expected generation** | Solar energy used by the hotel each year | 291,000 kWh/yr (24,250 kWh/month) |

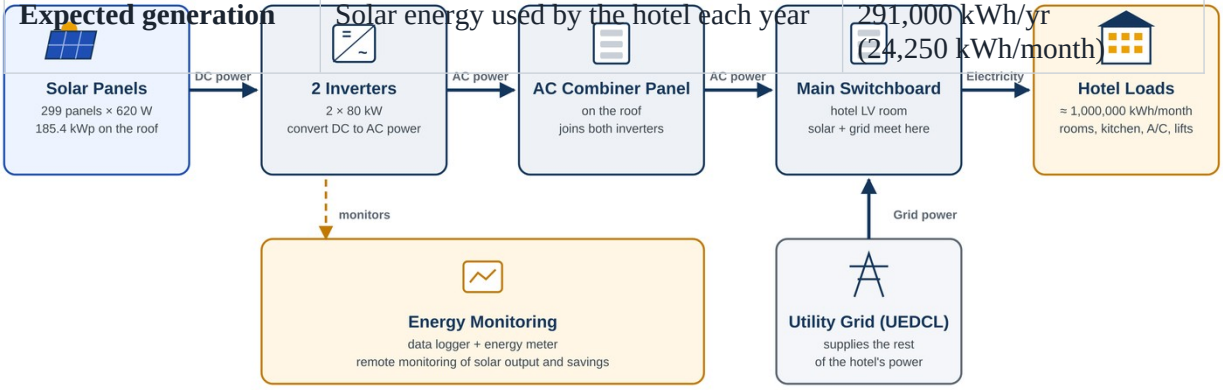



*Figure 4 : Roof Top PV System Block Diagram*

## 7. Cost Estimate (Bill of Quantities)

| No. | Description | Qty | Rate (USD) | Amount (USD) |
|---|---|---|---|---|
| 1 | Supply of PV modules, 620 Wp (299 Nos Modules) | 185.4 kWp | 0.186 | 34,565 |
| 2 | Supply of String inverters, 80 kW | 2 nos | 4,000 | 8,000 |
| 3 | Supply of Mounting structure per Wp | 185.4 kWp | 0.095 | 17,634 |
| 4 | Supply of DC system | Lot | USD 0.031/Wp | 5,747 |
| 5 | Supply of AC system | Lot | Lump sum | 3,640 |
| 6 | Earthing, equipotential bonding & lightning protection integration | Lot | Lump sum | 1,847 |
| 7 | Installation, testing & commissioning | Lot | Lump sum | 15,466 |
| 8 | Transport, logistics & customs clearing | Lot | Lump sum | 10,150 |
| | Sub-total | | | 97,049 |
| | **Contingency** | | 10 % | **7,612** |
| | **Total Project Value (excluding taxes)** | | | **104,661** |
| | Specific cost | | | 0.565 USD/Wp |

*Table 7: Project Cost Estimation*

## 8. Financial Analysis

The solar system needs a total investment of USD 104,661 (total project value, excluding taxes). At the avoided shoulder tariff of USD 0.0871/kWh (UGX 357/kWh as billed, excl. VAT, at UGX 4,100/USD), rising 2% a year, it saves about USD 25.3k in its first year (PVsyst P50 generation).
The investment pays back in about 4.1 years. Over 25 years the project gives an IRR of 25.5% and an NPV of USD 153,091 at a 10% discount rate, with total savings of about USD 763,000.

Solar energy costs about USD 0.040/kWh, below the grid tariff, so every unit generated reduces the hotel's electricity cost. No O&M or inverter-replacement cost is included in the cash flow.

| | |
|---|---|
| Total investment (excluding taxes) | USD 104,661 |
| Avoided grid tariff (year 1) / escalation | USD 0.0871/kWh (shoulder) / 2 % per year |
| Year-1 savings | USD 25,343 |
| Simple payback | 4.1 years |
| Project IRR (25 years, unlevered) | 25.5 % |
| NPV at 10 % discount rate | USD 153,091 |
| Levelized cost of energy (LCOE) | USD 0.041/kWh |
| 25-year cumulative savings | USD 763,309 |

*Table 8 : Financial Analysis*

### Tariff sensitivity

The base case uses the shoulder tariff from the October 2026 bill, UGX 357/kWh (USD 0.0871/kWh at UGX 4,100/USD). If the 40 % energy rebate shown on the bill is applied to the saved energy, the avoided cost falls to USD 0.0522/kWh; if VAT on electricity cannot be recovered it rises to USD 0.1027/kWh. Each USD 0.01/kWh change in the avoided tariff moves the NPV by about USD 29.6k. The break-even tariff (NPV = 0 at 10 %) is about USD 0.035/kWh, roughly 59 % below the current shoulder tariff.

| Avoided tariff (USD/kWh) | Payback (yrs) | IRR | NPV 10 % (USD) |
|---|---|---|---|
| 0.0522 (with 40 % rebate) | 6.6 | 15.4% | 49,990 |
| 0.0700 | 5.0 | 20.6% | 102,552 |
| 0.0871 (base – bill tariff) | 4.1 | 25.5% | 153,091 |
| 0.1000 | 3.5 | 29.1% | 191,357 |
| 0.1027 (VAT not recoverable) | 3.5 | 29.9% | 199,487 |

*Table 9: Tariff Sensitivity*

### Annual cash flow (selected years, USD)

For an investment of USD 104,661, the system saves about USD 25.3k in its first year, rising with the tariff*. The investment is recovered in about 4.1 years, and over 25 years the hotel's cumulative net cash flow reaches USD 658,648.

| Year | Generation (kWh) | Tariff (USD/kWh) | Savings | Net cash flow | Cumulative |
|---|---|---|---|---|---|
| 0 | | | | (104,661) | (104,661) |
| 1 | 291,051 | 0.087 | 25,343 | 25,343 | (79,319) |
| 2 | 286,988 | 0.089 | 25,489 | 25,489 | (53,830) |
| 3 | 285,840 | 0.091 | 25,895 | 25,895 | (27,935) |
| 4 | 284,697 | 0.092 | 26,307 | 26,307 | (1,629) |
| 5 | 283,558 | 0.094 | 26,726 | 26,726 | 25,097 |
| **6** | **282,424** | **0.096** | **27,151** | **27,151** | **52,248** |
| **7** | **281,294** | **0.098** | **27,583** | **27,583** | **79,831** |
| **8** | **280,169** | **0.100** | **28,022** | **28,022** | **107,854** |
| **9** | **279,048** | **0.102** | **28,469** | **28,469** | **136,322** |
| **10** | **277,932** | **0.104** | **28,922** | **28,922** | **165,244** |
| **12** | **275,713** | **0.108** | **29,850** | **29,850** | **224,476** |
| **15** | **272,418** | **0.115** | **31,298** | **31,298** | **316,907** |
| **18** | **269,162** | **0.122** | **32,817** | **32,817** | **413,824** |
| **20** | **267,013** | **0.127** | **33,870** | **33,870** | **481,034** |
| **25** | **261,715** | **0.140** | **36,654** | **36,654** | **658,648** |

*Table 10: Annual Cash Flow*

**Grid price saved for each kWh of solar. Starts at USD 0.087 (UGX 357/kWh shoulder tariff) and rises 2% a year to about USD 0.140 by year 25.*

## 9. Expansion Options

The target of about 500 kWp exceeds what the architect's solar zone on the mezzanine roof can hold. Because the hotel load is around ten times the PV output, every additional kWp would also be fully self-consumed, so expansion can be added in stages without a battery.

| Option | Capacity | Year-1 energy | Budget cost (USD) | Remarks |
|---|---|---|---|---|
| Phase 1 – mezzanine roof (this report) | 185.4 kWp | 291 MWh | 104,661 | Ready for detailed design |
| Phase 2 – car-park solar canopies | 250 kWp | 370 MWh | 187,500 | Adds shaded parking; steel canopy cost |
| Indicative total | 435 kWp | 661 MWh | 292,161 | 5.6 % of hotel consumption |

## 10. Assumptions, Exclusions and Next Steps

### Key assumptions

- The hotel used 981,901 kWh in October 2026 (UEDCL bill) with an average daytime demand of about 1,348 kW – far above the 160 kW PV output.
- The avoided tariff is the UEDCL shoulder tariff as billed, UGX 357/kWh ≈ USD 0.0871/kWh at UGX 4,100/USD (excl. VAT), escalating 2 %/yr. The bill also shows a 40 % energy rebate; if it is applied to the solar savings the value is USD 0.0522/kWh.
- Prices are 2026 East African market levels in USD, excluding taxes.
- Energy yield from the PVsyst V8.1.8 simulation, P50 as simulated (291,051 kWh/yr); shading and soiling are not yet modelled and a P90 year is about 8 % lower.

### Exclusions

- Battery storage, diesel-generator synchronisation and export metering.
- Roof waterproofing renewal, structural strengthening, and civil works to the LV room.
- Utility network upgrades, permit fees and financing costs.

Annexes

1.Single Line Diagram
2.Layout Drawing
3.Data Sheets of Inverter
4.Data Sheet of Solar Module
5.Stuctrural Drawing
6.Quotation
7.PVsyst Simulation Report